Jack Shi

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Education

University of California Los Angeles — Los Angeles, CA
B.S. in Electrical Engineering — September 2024 – June 2028
GPA: 3.91/4.00

- Tutoring Lead, Tau Beta Pi Engineering Honor Society,

Work Experience

Archer Aviation — San Jose, CA
Cell Test Engineer Intern — June 2026 – August 2026

- Build Python services unifying all cyclers behind one vendor-agnostic API, automating schedule deployment, closed-loop cell thermal control, & telemetry across hundreds of lithium-ion test channels.
- Owned MCU firmware, HTTP servers, and interface contracts for eVTOL cell test towers; built Python-controlled HIL benches with Analog Discovery 2 to emulate hardware signals, inject sensor faults, validate board behavior and safe-state responses, and support FMEA.
- Diagnosed excess Peltier heating caused by a motor driver mismatched to a resistive load; designed an alternative drive approach using increased PWM frequency and an LC filter, reducing simulated RMS ripple by 28% and power draw 49% and achieving an 18°C temperature differential in experimental testing.

UCLA – Emergence of Communication Lab, UC San Diego *GitHub Link* — Los Angeles, CA
Lab Manager, Mixed-Signal Embedded Systems — November 2025 – Present

- Recruited and lead a 4-person engineering team developing a mixed-signal wearable sensing system across UCLA and UCSD collaborations, defining architecture, milestones, and hardware/firmware integration.
- Architect a custom four-layer mixed-signal PCB integrating an ESP32-S3, USB power/data, microSD, camera, and MEMS/contact microphones, consolidating distributed modules into a compact platform.
- Designed a piezoelectric contact-microphone analog front end (op-amp buffer, filtering, ferrite-bead supply isolation) to address high source impedance, validating through SPICE and bench testing; wearable selected for UCLA patent application, with recognition as Lead Inventor.

University of California San Diego – Design Lab ECE Branch — San Diego, CA
R&D Embedded Systems Intern – OpenBabyographer Project — April 2025 – September 2025

- Embedded System Integration – Developed a wearable embedded sensing system using Raspberry Pi, integrating camera, MEMS microphone, & power management hardware; contributed to a 160-gram system deployed across 10 infant sessions and to the hardware section of the resulting manuscript.
- Hardware–Software Co-Design – Assembled & soldered custom circuitry; designed snap-fit enclosures in Fusion 360 & iteratively refined sensor placement to ensure robustness, durability, & stable operation.

Featured Projects

Analog UV Exposure Wearable — October 2026 – Present

- Designed, simulated, and prototyped a coin-cell-powered, fully analog UV exposure wearable, integrating a photodiode transimpedance amplifier and precision op-amp signal conditioning.
- Implemented active voltage clamps, an analog dose integrator, and closed-loop MOSFET current drivers for continuous UV-dependent RGB indication and cumulative exposure alerts without a microcontroller.

Tabquarium – ESP32-C3 Embedded Aquarium Game *GitHub Link* — March 2026 – June 2026

- Engineered an ultra-compact ESP32-C3 handheld gaming device with a custom PCB and CAD-designed mechanical soda-tab scanning interface for physical item collection.
- Developed event-driven embedded firmware with persistent game-state storage, low-power sleep/wake management, and an interactive I^2C OLED GUI featuring menus and aquarium simulation.

Dining Buddy Smart Keychain Project *GitHub Link* — May 2025 – October 2025

- Designed and fabricated an ultra-compact ESP32-C3 smart keychain integrating a custom PCB, 3D-printed snap-fit enclosure, and gesture-controlled single-button OLED interface.
- Developed embedded C++ firmware implementing HTTP menu/weather retrieval, SPIFFS flash caching, gesture-based navigation, and scheduled wake-ups with deep-sleep power optimization.

Papers

Tang, Liu, Shirazi, Hennessy, **Shi**, Bhamidipati, Nambiar, Luo, Deak, Wang — San Diego, CA
OpenBabyographer: A Multimodal Open-Source Infant Wearable for Studying Naturalistic Interactions from an Egocentric Perspective *Paper Link (Preprint / Technical Report)*

Skills

- **Hardware Equipment:** Oscilloscopes, Multimeters (DMM), Signal Generators, Power Supplies
- **Software Tools:** KiCad, Fusion360, LTspice, Git, Postman, React Native
- **Programming Languages:** Embedded C/C++ (RTOS), Python, MATLAB, SQL